

Micro-macro changepoint inference for periodic data sequences

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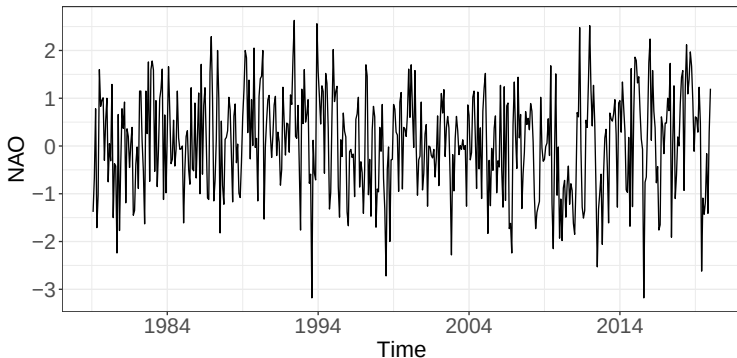
UCLA, Feb 2025

- Motivation
- Micro changepoint model
 - Bayesian Implementation
 - Frequentist Implementation
- Macro changepoint model
- Simulations
- Application

Motivation

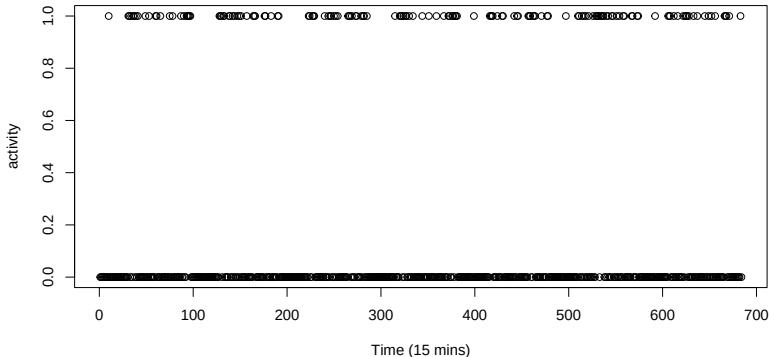
- How can we make the most of highly granular time series?
- How can we study changes in periodic and cyclical behaviours?

NAO fluctuations (1979–2020)



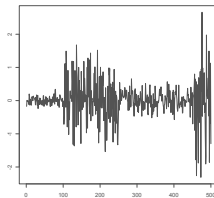
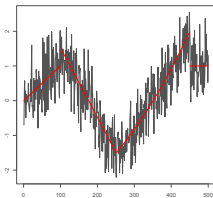
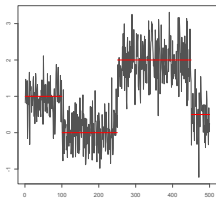


- How can we study changes in periodic and cyclical behaviours?



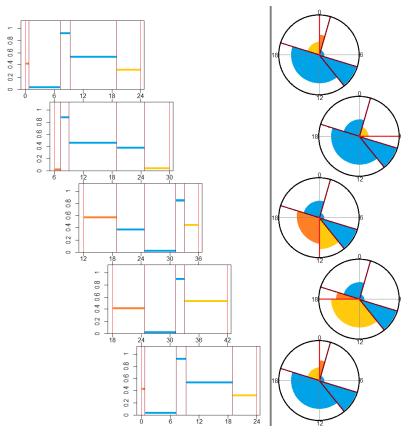
What are changepoints?

For data $y_1, \dots, y_n$, a changepoint is a location τ where the statistical properties of $y_1, \dots, y_\tau$ are different from $y_{\tau+1}, \dots, y_n$ in some way. Traditional changepoints include: mean, regression, variance.



- Binary segmentation & PELT
 - Assumes conditional independence ...
 - ... but dependence exists across days.
- Multivariate time series
 - Stack days to align changepoints ...
 - ... but what should 'time zero' be?

- Circular changepoint
 - For circular data (e.g. direction)
...
 - ... but time axis is linear.
- Circular Binary Segmentation
 - local time axis is circular ...
 - ... but just joins up ends not each day.
- Cyclic-HMM – Achieves periodicity via a structured transition matrix ... but period duration is stochastic.



Create method for periodic and global changepoint detection

- Wide range of data structures (assume likelihood available)
- Present Bayesian (RJMCMC fit) and frequentist for periodic
- Use PELT to extend to periodic

Create layered visualizations for ease of inference.

We denote changepoints as micro (periodic) and macro (global).

Periodic level changepoint detection

Challenge: How to incorporate circular nature

Soution: Reframe $t = cN + i$, $c \in \mathbb{N}_0$ and $i \in \{0, 1, \dots, N - 1\}$

Challenge: How to detect using (c, i) instead of t ?

Challenge: There is no single changepoint setting anymore!

- $\{X_t\}_{t \geq 0}$ — Real-valued time series.
- $N \in \mathbb{N}$ — Given period length.
 - $t = cN + i$ where $c \in \mathbb{N}_0$ and $i \in \{0, 1, \dots, N - 1\}$.
- $m \geq 1$ — Number of segments partitioning the period.
- $\tau_1, \dots, \tau_m \in \{0, 1, \dots, N - 1\}$ — Changepoint positions.
 - Ordered: $\tau_{j-1} < \tau_j$.
 - Periodic: $\tau_{m+1} = \tau_1$.
 - Minimum separation: $\tau_j - \tau_{j-1} \geq l$ for $l > 0$.
- μ_j for $j \in \{1, \dots, m\}$ — Segment means.
- σ_j^2 for $j \in \{1, \dots, m\}$ — Segment variances.

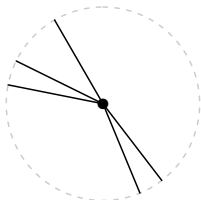
$X_t \sim \text{Normal}(\mu(t), \sigma^2(t))$ for $t \geq 0$ where
 $\mu(t) = \sum_{j=1}^m \mu_j \mathbb{I}\{i \in (\tau_{j-1}, \tau_j]\}$ and $\sigma^2(t) = \sum_{j=1}^m \sigma_j^2 \mathbb{I}\{i \in (\tau_{j-1}, \tau_j]\}$.

Bayesian

- $m \sim \text{Truncated - Poisson} \left(1, \{1, \dots, \lfloor \frac{N}{T} \rfloor\} \right)$.
- $\mu_j | \sigma_j^2, \tau, m \sim \text{Normal}(d, c\sigma_j^2)$ for $j = 1, \dots, m$.
- $\sigma_j^2 | \tau, m \sim \text{InvGamma}(a, b)$ for $j = 1, \dots, m$.
- Transform τ to excess segment length:
 - $\delta_j = \tau_j - \tau_{j-1} - l$ for $j = 1, \dots, m$.
 - $(\delta_1, \dots, \delta_m) \sim \text{Dirichlet - Multinomial}(N - lm, \alpha \mathbf{1}_m)$
- $\tau_1 \sim \text{Uniform}\{0, \dots, N - 1\}$.

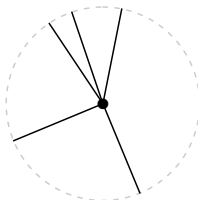
Cluster

$\alpha = 0.5$



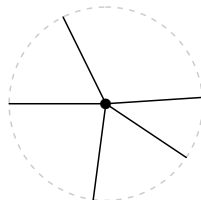
Random

$\alpha = 1$



Repel

$\alpha = 2$



Segment probabilities: Posteriors conditional on a sampled number and location of changepoints are easy to compute via conjugacy.

Number of segments and changepoint positions:

$$\pi(\boldsymbol{\tau}, m | \mathbf{x}) \propto \begin{cases} B(1 + \sum_t x_t, 1 + \sum_t (1 - x_t)) & m = 1, \\ \prod_{j=1}^m \left[\frac{B(1+s_j, 1+n_j-s_j)}{(\alpha+\delta_j)B(\alpha, \delta_j+1)} \right] \frac{(m\alpha+N-lm)B(m\alpha, N-lm+1)}{m!N} & \text{o/w,} \end{cases}$$

where $B(\cdot, \cdot)$ is the beta function.

- Trivial to evaluate $\pi(\boldsymbol{\tau}, m | \mathbf{x})$ for a given changepoint vector.
- Support size is potentially impractically large:
 - For $N = 96$ and $l = 4$, then there are $\sim 27.3 \times 10^{12}$ events.

- Permits exploration of varying dimensional spaces.
- Need to satisfy ‘dimension-matching’ condition.
- Focus inference on the marginal posterior $\pi(\boldsymbol{\tau}, m | \mathbf{x})$.

Algorithm

1. Sample a segment $j \in \{1, \dots, m\}$ uniformly.
2. Define the proposal set $\mathcal{P}_j(\boldsymbol{\tau}, m)$.
3. Sample a changepoint vector from the proposal pmf:

$$q(\boldsymbol{\tau}', m' | \boldsymbol{\tau}, m) \propto \pi(\boldsymbol{\tau}', m' | \mathbf{x}) \mathbb{I} \{ (\boldsymbol{\tau}', m') \in \mathcal{P}_j(\boldsymbol{\tau}, m) \}.$$

4. Accept/Reject proposal. Return to 1.

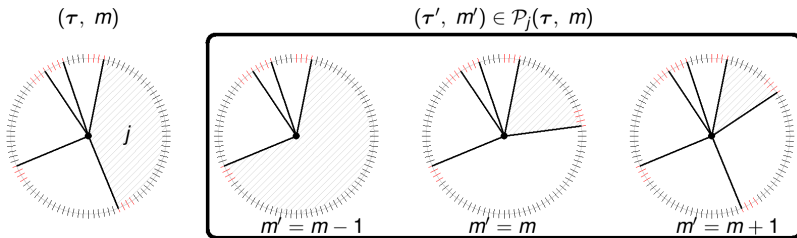
- Evaluate $\pi(\phi | \boldsymbol{\tau}, m, \mathbf{x})$ given the changepoint posterior samples.

Proposal set, $\mathcal{P}_j(\tau, m)$, contains three move types:

- Combine sampled segment with the next.
- Contract or extend the length of the current segment.
- Partition current segment into two.

Unique mapping

For each $(\tau', m') \in \mathcal{P}_j(\tau, m)$ excluding (τ, m) , there exists a unique segment j' such that $(\tau, m) \in \mathcal{P}_{j'}(\tau', m')$.



Frequentist

Challenge: No single changepoint setting to build upon.
Standard:

$$\begin{aligned}c_{0,t}^m &= \min_{\tau_1, \dots, \tau_m} \sum_{j=1}^{m+1} \mathcal{C}(x_{(\tau_{j-1}+1): \tau_j}) \\ &= \min_{\tau_{m-1}} [c_{0, \tau_{m-1}}^{m-1} + \mathcal{C}(x_{(\tau_{m-1}+1): t})].\end{aligned}$$

Circular:

$$\begin{aligned}c_{0,t}^2 &= \min_{\tau_1, \tau_2} [\mathcal{C}(x_{(\tau_1+1): \tau_2}) + \mathcal{C}(x_{(\tau_2+1): \tau_1})] \\ c_{0,t}^m &= \min_{\tau_{m-1}} [c_{0, \tau_{m-1}}^{m-1} + \mathcal{C}(x_{(\tau_{m-1}+1): t})] \quad m = 3, \dots, M\end{aligned}$$

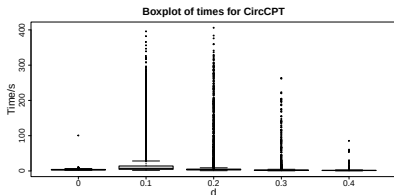
Correct detection ($1-\alpha$ and power rates)

$d \backslash \tau_1$	4	8	16	24	32	48
0.0	1.000	0.998	1.000	1.000	0.999	0.999
0.1	0.003	0.009	0.073	0.130	0.134	0.163
0.2	0.365	0.516	0.544	0.535	0.514	0.552
0.3	0.891	0.841	0.786	0.806	0.826	0.827
0.4	0.970	0.951	0.897	0.917	0.912	0.915

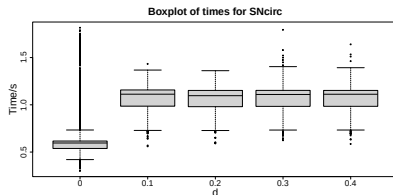
Correct detection ($1-\alpha$ and power rates)

$d \backslash \tau_1$	4	8	16	24	32	48
0.0	1.000	1.000	1.000	1.000	1.000	1.000
0.1	0.000	0.004	0.042	0.102	0.123	0.156
0.2	0.177	0.467	0.550	0.540	0.519	0.570
0.3	0.842	0.846	0.828	0.856	0.866	0.873
0.4	0.973	0.975	0.971	0.972	0.966	0.976

Simulation Results

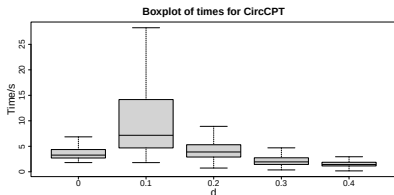


Bayesian

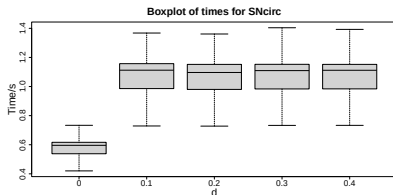


Frequentist

Simulation Results



Bayesian



Frequentist

Macro Level Changepoint Detection

- Treat the micro-level detection as a “fit” for a segment ...
- ... thus we can use traditional, linear time algorithms.

We use PELT which optimizes:

$$\sum_{i=1}^{q+1} [\mathcal{C}(y_{(\alpha_{i-1}+1):\alpha_i})] + \beta q. \quad (1)$$

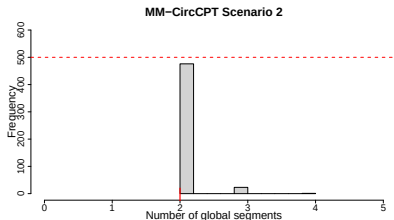
where q is the number of macro level changepoints and $\mathcal{C}(\cdot)$ is the (MAP) likelihood:

$$\mathcal{C}(y_{(\alpha_{i-1}+1):\alpha_i}) = -2 \sum_{j=1}^{\hat{m}_i} \sum_{t \in B_{i,j}^*} \log f(y_t | \hat{\mu}_{i,j}, \hat{\sigma}_{i,j}^2)$$

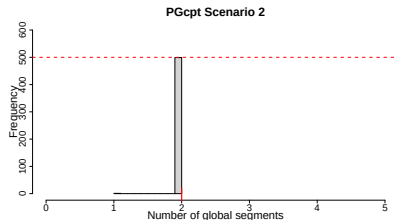
Simulations

See pre-print for full simulation study.

Simulation Results



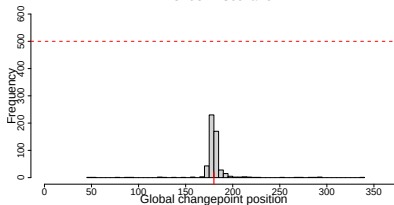
Bayesian



Frequentist

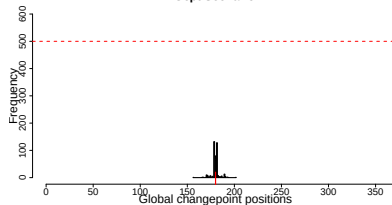
Simulation Results

MM-CircCPT Scenario 2



Bayesian

PGcpt Scenario 2

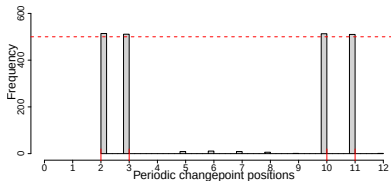


Frequentist

Simulation Results

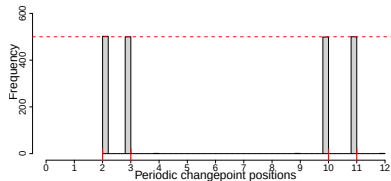


MM-CircCPT Scenario 2



Bayesian

PGcpt Scenario 2



Frequentist

Application



NAO fluctuations (1979–2019)

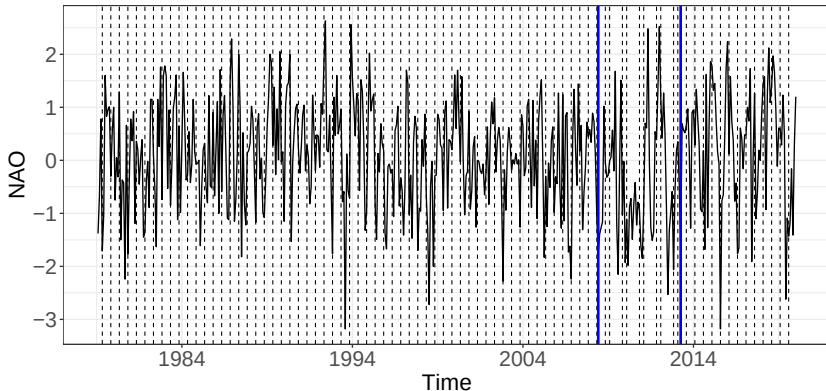


Figure: NAO data and Changepoint positions using $pen = 4 \log(n)$

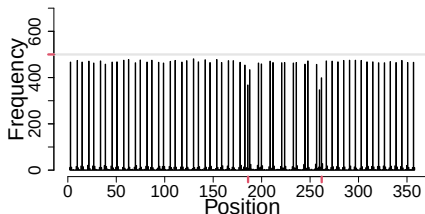
Macro-level segment	Micro-level changepoints
1979-2008	April and October
2008-2013	May and August
2013-2019	April and October

Table: Macro- and micro-level changepoint positions for the NAO index.

Overview:

- A new approach for identifying within-period changepoint events for time series.
Taylor, S. A. C., Killick, R., Burr, J. and Rogerson, L. (2021)
[‘Changepoint Detection within Periodic Binary Time Series’](#). JRSS:Series C.
- Pooling of evidence across multiple periods by applying a circular perspective to time.
- Introduced the micro-macro level changepoint approach
Ushakova, A., Taylor, S. A. C., Killick, R. (2023)
[‘Micro-Macro Changepoint Inference for Periodic Data Sequences’](#). JCGS.
- Bayesian and Frequentist viewpoints for a wide set of data types
Li, O., Killick, R. (2025+)
[‘Detecting changes in periodic data’](#). Submitted

Scenario 11



Scenario 7

